

RW-Post: Auditable Evidence-Grounded Multimodal Fact-Checking in the Wild

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Abstract

*Multimodal misinformation increasingly leverages visual persuasion, where repurposed or manipulated images strengthen misleading text. We introduce **RW-Post**, a post-aligned **text-image benchmark** for real-world multimodal fact-checking with auditable annotations: each instance links the original social-media post with reasoning traces and explicitly linked evidence items derived from human fact-check articles via an LLM-assisted extraction-and-auditing pipeline. RW-Post supports controlled evaluation across closed-book, evidence-bounded, and open-web regimes, enabling systematic diagnosis of visual grounding and evidence utilization. We provide **AgentFact** as a reference verification baseline and benchmark strong open-source LVLMs under unified protocols. Experiments show substantial headroom: current models struggle with faithful evidence grounding, while evidence-bounded evaluation improves both accuracy and faithfulness. Code and dataset will be released at <https://github.com/xudanni0927/AgentFact>.*

1. Introduction

The proliferation of social media has accelerated the spread of persuasive multimodal misinformation, where images and videos are used to shape perception alongside misleading text, posing risks to public health and social behavior [2, 7, 8]. Recent advances in generative AI further lower the cost of producing realistic deceptive content [1, 12, 38, 46]. These challenges motivate the need for automated systems capable of verifying multimodal claims with reliable evidence.

Existing approaches to multimodal misinformation detection often focus on narrow subtasks, such as out-of-context image detection or deepfake detection [13, 21, 32, 33, 37]. More recent work leverages LLMs and LVLMs with chain-of-thought prompting, claim decomposition, or retrieval augmentation for fact-checking [9, 18, 20, 29, 40, 44]. However, these systems frequently rely on shallow evidence usage and lack faithful grounding in verifiable

sources. Progress is further limited by the absence of benchmarks that jointly provide real-world multimodal claims, auditable reasoning traces, and explicitly linked evidence.

To address this gap, we introduce **RW-Post**, a **text-image benchmark** for real-world multimodal fact-checking. RW-Post pairs fact-checked claims with their original social-media posts, preserving contextual cues critical for verification. The dataset additionally provides reasoning traces and explicitly linked evidence extracted from human-written fact-checking articles through an LLM-assisted extraction-and-auditing pipeline. RW-Post supports both **closed-book** verification using only the post context, **evidence-bounded** verification using oracle evidence in dataset, and **open-web** verification with external evidence retrieval. To ensure fair evaluation in the open-web setting, we standardize the search backend and apply strict anti-leakage filtering.

To establish reference baselines, we introduce **AgentFact**, a modular verification agent-based framework that integrates strategy planning, evidence retrieval, visual analysis, reasoning, and evidence-grounded explanation generation, combined by retrieval-reasoning iterated workflow. We benchmark representative open-source LVLMs under unified prompting protocols and controlled evaluation settings. Experiments demonstrate that RW-Post reveals substantial headroom for multimodal verification: even strong baselines struggle with faithful evidence grounding and visually informed reasoning, highlighting promising directions for future research.

Our contributions are:

1. We introduce **RW-Post**, a benchmark for explainable multimodal fact-checking with post-aligned claims, reasoning traces, and source-grounded evidence.
2. We propose an LLM-assisted pipeline for extracting and auditing reasoning and evidence from human-written fact-checking articles.
3. We provide **AgentFact** and benchmark strong LVLMs under controlled closed-book, evidence-bounded, and open-web evaluation protocols.

2. Related Work

2.1. Misinformation and OOC Detection Datasets

Misinformation datasets can be broadly grouped into three categories: (1) *collected datasets*, (2) *synthetic datasets*, and (3) *OOC-specific datasets*, as summarized in Table 1.

Collected datasets. Early datasets compile verified claims from fact-checking websites such as PolitiFact and Snopes (e.g., FakeNewsNet [35], LIAR-RAW [41], MuMiN [28]). Many are text-only and suffer from noisy annotations or missing context, since social-media posts are often matched to claims via keyword or similarity heuristics. Large-scale corpora such as Fakeddit [26] further rely on weak labels from platform tags. AVERITEC [34] improves annotation quality by extracting explanations from fact-checking articles, but remains limited to text-only claims.

Recent efforts introduce multimodal resources. CLAIMREVIEW+ manually links images to fact-checking claims, while Mocheg heuristically extracts rationales from Snopes articles. However, these datasets often lack original posts, reliable evidence links, or consistent explanations.

Synthetic and OOC datasets. Synthetic datasets increase scale by generating claims from structured sources such as Wikipedia (e.g., FEVER, FEVEROUS, HOVER) or LLM-generated content. However, they mainly focus on textual entailment and rarely capture multimodal interactions. OOC datasets instead examine mismatches between real images and misleading text, using random mismatches (e.g., MAIM, COSMOS [6, 15]), semantic mismatches (e.g., NewsCLIPPings [23]), or entity substitution [30]. Despite these advances, synthetic and OOC datasets often lack realistic post context and interpretable explanations.

These limitations highlight the need for a realistic multimodal fact-checking benchmark that preserves *post-level context* and provides *auditable evidence*. **RW-Post** addresses this gap by aligning fact-checked claims with their original social-media posts and linking them to reasoning traces and verifiable evidence.

2.2. Misinformation and OOC Detection Methods

Existing approaches can be categorized along two axes: (1) internal estimation vs. external retrieval, and (2) feature-based vs. LLM/LVLM-based reasoning.

Feature-based internal methods rely on linguistic style, propagation patterns, or image–text consistency signals [23, 31, 36, 45]. However, these approaches lack external grounding and often struggle with real-world variability.

Feature-based external approaches retrieve related captions or images and compare them using semantic or entity-level similarity [3, 6, 25]. While retrieval improves robustness, these methods typically rely on shallow matching.

Recent **LLM/LVLM-based** systems leverage prompting and retrieval augmentation, sometimes in agent-style pipelines [9, 29, 40], improving interpretability but still exhibiting superficial evidence use in complex multimodal verification.

A central challenge is **faithfulness**: models can generate fluent rationales while hallucinating reasoning or misusing evidence, where explanations are not supported by verifiable sources. This has motivated evidence-grounded evaluation protocols that require citing evidence and auditing whether citations truly justify predictions, especially in open-web settings where retrieval adds variability and leakage risks. These limitations highlight the need for benchmarks that provide explicitly linked evidence and reasoning traces and support controlled regimes (e.g., closed-book vs. evidence-bounded vs. open-web) for systematic auditing and diagnosis.

Table 1. Fact-checking and out-of-context (OOC) datasets. *I*: image; *C*: claim; *P*: post; *R*: reasoning traces; *E_{src}*: external evidence sources; *E_{link}*: reasoning–evidence alignment for auditing. ✓/✗ denote provided/not provided.

Dataset	<i>I</i>	<i>C</i>	<i>P</i>	<i>R</i>	<i>E_{src}</i>	<i>E_{link}</i>	Size
<i>Synthetic Fact-checking Datasets</i>							
HOVER [16]	✗	✓	✗	✗	✓	✗	26k
FEVEROUS [5]	✗	✓	✗	✗	✓	✗	87k
Multi-News FC [10]	✓	✓	✗	✗	✓	✗	1291k
<i>OOC-specific Datasets</i>							
MMFakeBench [22]	✓	✗	✓	✗	✗	✗	11k
MAIM [15]	✓	✗	✓	✗	✗	✗	239k
COSMOS [6]	✓	✗	✓	✗	✗	✗	453k
NewsCLIPPings [23]	✓	✗	✓	✗	✗	✗	988k
<i>Collected Fact-checking Datasets</i>							
Twitter [17]	✓	✗	✓	✗	✗	✗	13k+
Weibo [17]	✓	✗	✓	✗	✗	✗	9k+
FakeNewsNet [35]	✓	✓	✓	✗	✗	✗	19k+
Fakeddit [26]	✓	✗	✓	✗	✗	✗	1M+
Weibo21 [27]	✓	✗	✓	✗	✗	✗	9k+
MuMiN [28]	✓	✓	✓	✗	✗	✗	21M+
RAWFC [41]	✗	✓	✗	✓	✓	✗	2k
LIAR-RAW [41]	✗	✓	✗	✓	✓	✗	12k
AVERITEC [34]	✗	✓	✓	✓	✓	✗	4.57k
Mocheg [42]	✓	✓	✗	✓	✓	✗	15k
Factify 2 [24]	✓	✓	✓	✗	✓	✗	50k
MR ² [14]	✓	✓	✓	✗	✓	✗	14k+
CLAIMREVIEW+ [9]	✓	✓	✗	✗	✗	✗	0.3k
RW-Post (ours)	✓	✓	✓	✓	✓	✓	1.52k

3. RW-Post: A Post-aligned Multimodal Fact-Checking Benchmark with Auditable Evidence

We introduce RW-Post, a benchmark for explainable multimodal fact-checking that reflects real-world verification scenarios. Each claim is paired with its original social-media post, structured reasoning (logic and key points), and explicitly linked evidence. Instead of using loosely relevant documents, RW-Post emphasizes targeted evidence that directly supports factual reasoning, aligning with the rigor of professional fact-checking.

Benchmark Regimes. RW-Post supports three complementary evaluation regimes: (1) **closed-book**, where models observe only the post and image; (2) **evidence-bounded**, where models receive the post together with a fixed set of evidence items (with stable IDs and source URLs); and (3) **open-web**, where models retrieve evidence via web search under a standardized, leakage-controlled protocol. These regimes enable systematic analysis of visual grounding, evidence utilization, and explanation faithfulness.

Following common benchmark practice [11, 22, 43], RW-Post is released with a development and a test split (1:5), comprising 252 and 1,268 samples, respectively. We use the dev split for prompt and system development and report all benchmark results on the test split; while our main experiments are zero-shot, the dataset can also support supervised finetuning.

We collect 13,000 Snopes fact-checking articles (2017–2024) and apply strict filtering and quality control. To enable structured extraction, webpages are reformatted into a *text-with-tagged-URL* representation that preserves the contextual position of each link. This representation allows LLMs to reliably extract posts, images, reasoning, and evidence. The pipeline is illustrated in Figure 1.

Data collection. We collect fact-checking articles from Snopes, which covers diverse misinformation topics and follows professional verification standards. For each article, we extract the *URL*, *category*, *headline*, *claim*, *truthfulness label*, and *full HTML content*.

Text and image filtering. Articles shorter than 90 words or longer than 2,000 words are removed to avoid insufficient or overly verbose content. Images smaller than 200×200 pixels are also excluded.

3.1. Dataset annotation

Relabeling truthfulness labels. Snopes provides over ten fine-grained truthfulness labels, some of which have overlapping meanings (e.g., *MISCAPTION* vs. *FALSE*) and highly imbalanced distributions. To simplify evaluation and improve label consistency, we consolidate them into five high-level categories based on factual accuracy: *True*, *False*, *Unproven*, *Mixture*, and *Outdated*. Labels unrelated

to factual correctness (e.g., satire) are excluded. Table 2 shows the mapping from original Snopes labels to our consolidated categories.

Table 2. Mapping from original labels to high-level labels.

Label	Original Labels
False	FALSE, MISCAPTION, MISCAPTIONED, FAKE, LEGEND, SCAM, MISATTRIBUTED
Mixture	MOSTLY FALSE, MIXTURE, MOSTLY TRUE
Outdated	OUTDATED
True	TRUE, CORRECT ATTRIBUTION, LEGIT
Unproven	UNFOUNDED, UNPROVEN, RESEARCH IN PROGRESS

Article reformatting. Inspired by Flamingo [4], to enable structured extraction, we convert each HTML article into a *text-with-tagged-URLs* representation. All URLs in the webpage are identified and categorized as image, video, or webpage links based on file extensions, and then inserted back into the text with modality tags while preserving their original positions. This representation maintains the contextual relationship between text and links, allowing LLMs to reliably identify claim-aligned posts, images, and evidence, and aligning evidence with reasoning. Examples of this format are shown in Figure 1.

Explanation extraction. We design an LLM-assisted extraction pipeline to generate structured annotations from the reformatted articles. Using GPT-4o, we extract: (1) the original post and associated images, (2) a concise reasoning overview and itemized key points, and (3) evidence aligned to each key point with stable evidence IDs and URLs.

3.2. Quality Control

To ensure dataset quality, we apply both automated filtering and human verification.

Multimodal relevance filtering. We remove claims where the image does not play a meaningful role in verification. Specifically, GPT-based prompts determine whether the image serves as direct evidence, strengthens the claim’s credibility, or significantly affects its perceived truthfulness. Claims judged as not requiring image context are excluded.

Watermark leakage filtering. Images containing fact-checking watermarks (e.g., “fake”, “false”, or “misleading”) are removed to prevent label leakage.

Both steps combine GPT-based filtering with manual inspection. Manual checks show high consistency with the automated filtering results, indicating reliable identification of multimodal claims and watermark-free images. Prompt templates and detailed evaluation statistics are provided in Appendix .1.

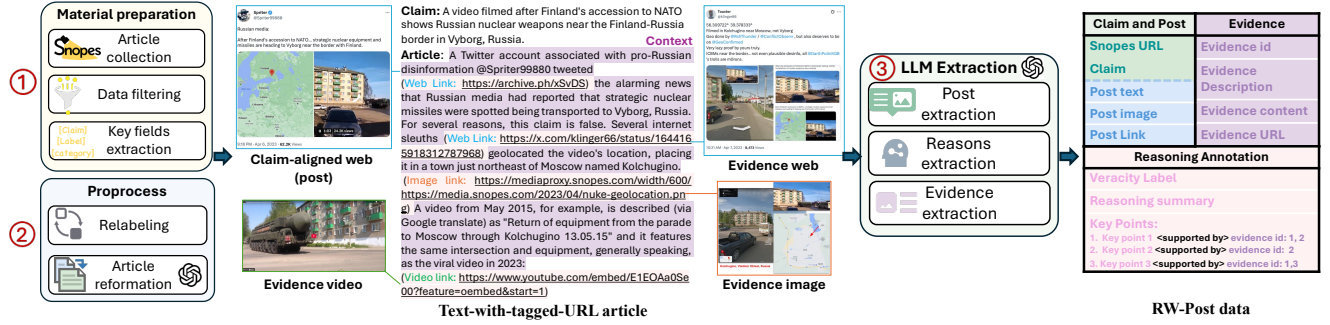


Figure 1. RW-Post Dataset: Use Context (purple highlight) helps LLM determine whether the link (pink highlight) is post or evidence and which rationale it supports.

3.3. Human validation

To verify the reliability of the LLM-extracted annotations, we conduct a human validation study on posts, images, reasoning, and evidence.

Post and Image Validation Annotators verify whether extracted posts and images correctly correspond to the original fact-checking articles. Each sample is independently annotated by two reviewers, with disagreements resolved through discussion. On a manually reviewed set of 100 articles, the system correctly extracts post links in 76.4% of link-containing cases, with most errors arising from missed extractions rather than incorrect matches. For images, annotators categorize their role in verification as *necessary*, *relevant*, or *irrelevant*. Results show that images provide essential evidence in 87% of claims, while only 8% correspond to cases with irrelevant image usage. Examples of these categories are shown in Figure 2.

Validation of Reasoning and Evidence We further validate the correctness of reasoning and evidence using the same rubric employed in our explainability evaluation (Sec. 5.4). Annotators assess whether (i) the reasoning logic and key points are supported by the linked evidence (reasoning hallucination), and (ii) the cited evidence is correctly used and accurately represented (evidence usage hallucination). Detailed guidelines and aggregated results are reported in Sec. 5.4 and Appendix 4.

3.4. Statistics

Summary statistics are shown in Figure 3. The RW-Post dataset spans 10 domains, with Fauxtography (831), Politics (384), Entertainment (107), Junk News (99), and Viral Phenomena (64) being the most frequent categories.

RW-Post's main benchmark split contains 1.52k instances with three labels (True/False/Unproven). We additionally provide a challenge set of 218 Mixture/Outdated

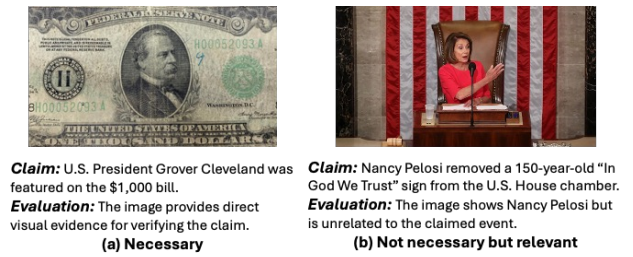


Figure 2. Examples of image annotations illustrating whether the image is *necessary* or *relevant* for claim verification.

cases for future research on nuanced and time-sensitive verification (Figure 3a); we do not include this set in the main benchmark metrics unless otherwise stated.

Most articles contain diverse modalities of evidence such as text, image, and multimodal combinations (Figure 3b), and mostly contain 3–8 pieces of evidence (Figure 3c). Categories like History, Health, and Viral Phenomena show higher median evidence counts (Figure 3d), indicating that human fact-checkers might conduct more in-depth investigations for these topics.

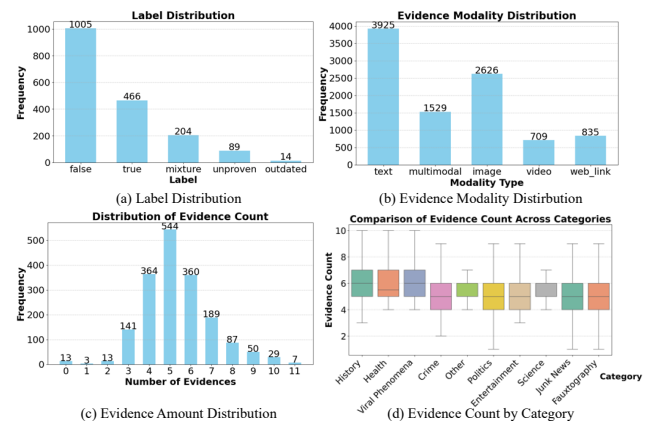


Figure 3. Statistics of RW-Post Dataset

4. Reference Verification Pipeline

Overview To facilitate reproducible benchmarking on RW-Post, we provide **AgentFact**, a reference multimodal verification pipeline. AgentFact decomposes open-web fact-checking into modular components for (i) strategy planning, (ii) textual evidence retrieval, (iii) visual analysis via reverse image search, and (iv) evidence-grounded reasoning and (v) explanation generation.

The pipeline supports both **open-web** evaluation (end-to-end evidence acquisition) and **evidence-bounded** evaluation (verification with fixed evidence), enabling systematic auditing of evidence usage and explanation faithfulness.

4.1. Problem Definition

A multimodal claim is represented as $\{C, P\}$, where C is the textual claim and $P = \{P_T, P_I\}$ denotes the post context consisting of post text and an associated image.

The goal is to determine the claim veracity and provide evidence-grounded reasoning. The system outputs:

- a veracity label V ,
- a set of evidence items $E = \{(id_i, x_i, u_i)\}$ containing evidence snippets and source URLs,
- a set of key reasoning points aligned with evidence K_A ,
- a reasoning summary explaining the verification process R_A ,
- and a confidence score C_L .

Evidence Representation. Evidence is represented as items with stable identifiers, $E = \{(id_i, x_i, u_i)\}$, where x_i is a short evidence description and u_i is the source URL. Reasoning points cite evidence using these identifiers, enabling explicit evidence grounding and auditing.

4.2. Agents for Fact-Checking

As illustrated in Figure 4, the framework consists of five specialized agents: 1) **Strategy Planning Agent (Agent-SP)**, 2) **Text Evidence Retrieval and Validation Agent (Agent-TR)**, 3) **Image Retrieval and Analysis Agent (Agent-IR)**, 4) **Reasoning Agent (Agent-R)**, 5) **Explanation Generation Agent (Agent-EG)**. Each agent is designed with task-specific knowledge to mimic distinct components of real-world fact-checking workflows, and is deployed using prompt-engineering techniques with tool interaction. The simplified prompts are provided in the Appendix .2.

4.2.1. Strategy Planning (Agent-SP)

Agent-SP takes as input the claim (C), the associated post (P), and—when applicable—previous reasoning trace. It generates a verification plan consisting of (i) reasoning steps S , (ii) a validation list L_v containing statements to verify, and (iii) a search list L_s describing auxiliary search intents.

4.2.2. Text Evidence Retrieval and Validation Agent (Agent-TR)

Agent-TR retrieves textual evidence from the web to support claim verification. The module operates through four steps:

1. **Query Generation.** Search queries Q are generated based on the search list L_s and validation list L_v produced by Agent-SP.
2. **Web Search.** The Serper API¹ is used to retrieve candidate webpages for each query.
3. **Query-Guided Filtering.** Retrieved results are filtered and summarized to retain only evidence relevant to the input queries.
4. **Source Reliability Estimation.** To reduce the influence of unreliable information, each source is classified into one of four categories: *reliable*, *unreliable*, *satire*, or *un-sure*. The resulting reliability-annotated evidence is denoted as E_r .

This design reduces the impact of misleading or satirical sources during reasoning.

4.2.3. Image Retrieval and Analysis Agent (Agent-IR)

Agent-IR analyzes the input image to detect potential visual inconsistencies in multimodal claims. The module first performs reverse image search using the Google Cloud Vision API to retrieve visually similar images and their associated webpage text.

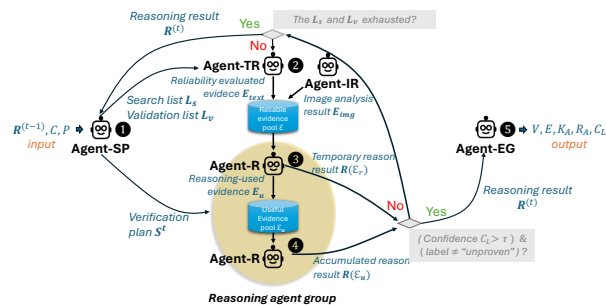
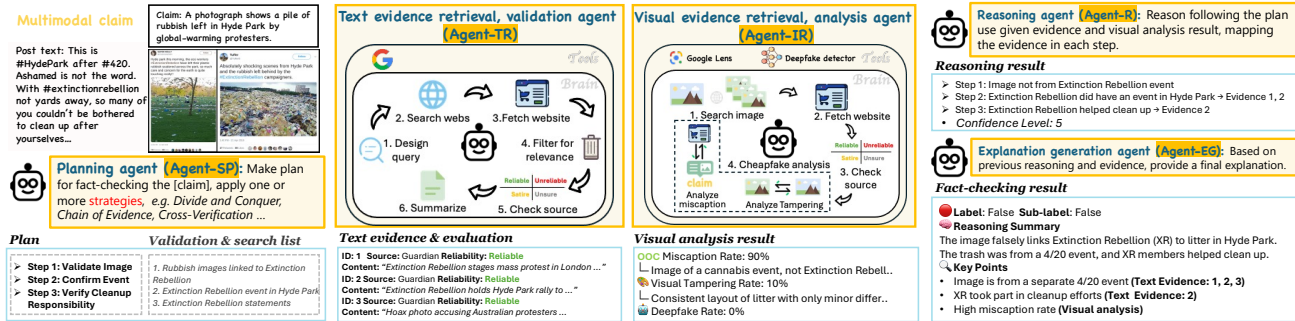
Retrieved images are categorized by an LLM into three types: (1) *near-duplicate images*, which are visually almost identical to the input image; (2) *same-event images*, which depict the same real-world event but differ in viewpoint or timing; (3) *unrelated images*, which are visually or semantically unrelated.

Images in the first two categories are retained as potentially informative evidence for verification. Based on these results, Agent-IR performs three types of verification: (i) **visual tampering detection**, by comparing the post image with retrieved images; (ii) **miscaption detection**, by checking consistency between the claim/post text and retrieved webpage text; (iii) **deepfake detection**, using the DIRE model [39]. The resulting signals are provided to the reasoning agent as auxiliary evidence.

4.2.4. Reasoning Agent (Agent-R)

The reasoning agent is provided with the claim (C), the post text (P_T), and the validation plan ($S^{(t)}$). It follows the prescribed reasoning steps, evaluating the relevant evidence at each step. Before the reasoning process begins, Agent-R is prompted to rephrase the input claim to improve focus and prevent the model from drifting away from the core verification target. The output (R) of Agent-R consists of

¹<https://serper.dev/>



three components: **Reasoning Results** — step-wise analysis, supporting evidence descriptions, and utility assessments; **Veracity Label** — the predicted truthfulness of the claim (e.g., true, false, unproven); **Confidence Level** — a score from 1 to 5 reflecting the model’s certainty.

Open-web Workflow As illustrated in Figure 5, open-web verification follows an iterative retrieve–reason loop:

tionally evaluate on Mocheg [42], CLAIMREVIEW2024+ (CR2024+) [9], and the OOC dataset NewsCLippings to test generalization [23]. Due to the cost of search-based methods, we sample 1,000 test instances for Mocheg and NewsCLippings, while using the full test sets for RW-Post and CR2024+. We follow each dataset’s official label space (binary for OOC and three-class for fact-checking).

Baselines. We compare AgentFact with three categories of baselines: (1) **feature-based OOC detectors** (CCN [3], Sniffer [32]), (2) **retrieval-augmented LLM/LVLM fact-checking systems** (LEMMA [40], DEFAME [9]), and (3) **open-source LVLMs** (LLaVA-1.5 and Qwen2-VL).

Evaluation Regimes. We evaluate models under the three benchmark regimes defined in Sec. 3: **closed-book**, **evidence-bounded**, and **open-web**. Search-based systems (AgentFact, LEMMA, and DEFAME) are evaluated in the open-web setting, while LVLMs are evaluated in the closed-book and evidence-bounded settings on RW-Post.

5.1. Implementation Details

Model and API Settings. For search-based systems (LEMMA, DEFAME, and AgentFact), we conduct an inference-only evaluation and standardize key components. We use GPT-4o-mini as the underlying LLM and the Serper API as the unified web search backend with a fixed top-10 results per query. LVLM baselines (LLaVA-1.5 and Qwen2-VL) are evaluated using their native models without additional LLM calls. For Sniffer, we use the publicly released pretrained checkpoints trained on NewsCLippings. For CCN, we report the results from the original paper, as reproducing the training requires access to the full dataset and preprocessing pipeline.

Computing Resources. Search-based systems are run on a CPU-only machine. LVLM baselines are evaluated on a single NVIDIA H100 GPU with temperature of 0 and a maximum output length of 256 tokens.

Evidence Settings. In evidence-bounded evaluation, models are provided with the linked RW-Post evidence items. In open-web evaluation, all retrieved URLs and evidence snippets are logged for auditing.

Data Leakage Prevention. To prevent leakage, we filter evidence by excluding (i) results from major fact-checking websites (e.g., Snopes; see Appendix .3) and (ii) sources published on or after the original fact-checking date.

5.2. Open-web End-to-End Veracity Results

Metrics We evaluate veracity prediction under two settings: **binary classification** for OOC detection and **three-class classification** (True, False, Unproven) for fact-checking. For both settings we report **weighted F1**, **accu-**

Table 3. Veracity classification results across diverse datasets. “OOC”: 2-class out-of-context. “FC”: 3-class fact-checking.

Dataset	Task	Model	F1	Acc	Rec	P
NewsCLippings	OOC	CCN	–	84.7	–	–
		Sniffer	86.5	86.5	86.5	86.5
		LEMMA	52.3	58.2	58.2	61.5
		DEFAME	60.7	60.8	60.8	60.7
		AgentFact	56.6	59.7	59.7	61.8
Mocheg	FC	LEMMA	28.0	37.5	37.5	40.6
		DEFAME	44.9	46.0	46.0	47.3
		AgentFact	51.9	48.2	48.2	49.6
ClaimReview	FC	LEMMA	16.5	14.5	14.5	87.3
		DEFAME	59.2	52.1	52.1	81.7
		AgentFact	69.2	67.0	67.0	74.6
RW-Post	FC	LEMMA	45.6	42.1	42.1	78.4
		DEFAME	51.3	46.5	46.5	80.1
		AgentFact	65.5	57.8	57.8	79.1

racy, weighted recall, and **weighted precision**. All metrics are reported as percentages (multiplied by 100).

Results. Table 3 reports end-to-end **open-web** veracity results under a unified search backend with leakage-controlled filtering. AgentFact achieves the best performance among fact-checking systems on the three fact-checking datasets, while specialized OOC detectors (CCN and Sniffer) remain strongest on the OOC task.

Fact-Checking (3-class). AgentFact consistently improves weighted F1 and recall over LEMMA and DEFAME on RW-Post and ClaimReview2024+, indicating that stronger evidence acquisition and evidence-grounded reasoning benefit complex real-world verification. On Mocheg, which lacks post-image context, AgentFact still achieves clear gains, suggesting that the pipeline generalizes to text-only verification.

OOC Detection (binary). On NewsCLippings, task-specific detectors outperform general-purpose fact-checking systems, as expected. This gap highlights the difference between OOC detection and open-web fact-checking, motivating the controlled RW-Post benchmark (Sec. 5.3) for deeper analysis of evidence utilization and visual grounding.

5.3. Controlled Benchmarking on RW-Post (LVLM Baselines)

To enable controlled and reproducible evaluation, we assess LVLMs on RW-Post under two regimes: (1) **closed-book**

Table 4. Controlled RW-Post evaluation. T: text; I: image; E: evidence.

Model	Setting	N	F1	Acc	Rec	P
<i>Qwen2-VL</i>	T	638	18.2	30.9	30.9	67.2
	T+I	638	18.7	30.3	30.3	71.9
	T+E	638	42.3	44.0	44.0	76.6
	T+I+E	638	35.5	40.8	40.8	75.6
<i>LLaVA-1.5</i>	T	1093	16.8	24.7	24.7	68.3
	T+I	1093	18.0	26.9	26.9	72.6
	T+E	1093	47.9	47.9	47.9	77.8
	T+I+E	1093	45.0	47.3	47.3	77.2
<i>AgentFact</i>	T+I+E	1267	86.3	85.5	85.5	87.3

and (2) **evidence-bounded**. We evaluate four input configurations: *T*, *T+I*, *T+E*, and *T+I+E*.

Results. Results are reported on the **intersection subset** of RW-Post instances successfully parsed under all four configurations (T, T+I, T+E, T+I+E), yielding 1,093 instances for LLaVA-1.5 and 638 for Qwen2-VL (Table 4).

Evidence substantially improves performance. Both LLaVA-1.5 and Qwen2-VL show large gains when evidence is provided (T→T+E). For example, Qwen2-VL improves from 18.2/30.9 (F1/Acc) to 42.3/44.0, and LLaVA-1.5 improves from 16.8/24.7 to 47.9/47.9. Therefore, explicit evidence is essential for accurate verification.

Visual grounding remains challenging. Adding image inputs provides little benefit in closed-book settings and slightly degrades evidence-bounded performance (T+E → T+I+E), suggesting that current LVLMs struggle to reliably integrate visual cues with textual evidence.

AgentFact under evidence-bounded evaluation. To isolate reasoning from retrieval, we evaluate AgentFact under a fixed evidence-bounded setting on RW-Post. It achieves the best performance (86.3 F1 / 85.5 Acc), indicating effective evidence-grounded reasoning.

5.4. Evidence Grounding and Explainability

We evaluate explanation quality through human assessment following prior LLM-based fact-checking studies [19, 44]. Forty RW-Post instances are randomly sampled by 3 annotators. Outputs from models and RW-Post labels are presented in randomized order with anonymized identifiers.

Annotation Scheme. Annotators evaluate each output along three dimensions: (1) **Reasoning Hallucination** (unsupported or logically invalid inference), (2) **Evidence Usage Hallucination** (misused, irrelevant, or fabricated evidence), and (3) **Label Justification** (whether the predicted label is well-supported). Each dimension is scored on a

three-level ordinal scale (0–2), where lower scores indicate fewer errors and higher faithfulness. For presentation, scores are normalized to the range [0, 1]. For outputs with explicit evidence citations (evidence IDs), annotators verify whether the cited evidence supports the stated key points and the predicted label. Detailed annotation guidelines and the interface are provided in Appendix 4.

Results. This human evaluation complements the evidence-bounded benchmark by auditing whether explanations are faithful to the provided evidence. Table 5 reports the average scores across three annotators. (1) Ground-truth rationales achieve near-perfect scores (0.94–1.00), demonstrating high faithfulness and serving as a strong upper bound for explanation quality. (2) AgentFact significantly outperforms LEMMA across all criteria. Using the Wilcoxon signed-rank test on paired differences, we obtain $p < 10^{-5}$ for reasoning, evidence usage, and label justification. Bootstrap confidence intervals (95%) further confirm the robustness of these improvements.

Table 5. Average human evaluation scores (0–1). Right columns report paired significance between AgentFact and LEMMA. Δ : mean paired difference (AgentFact – LEMMA).

Criteria	GT	LEMMA	AgentFact	Δ	p-value
Reasoning $\uparrow$	0.969	0.445	0.767	−0.65	6.8×10^{-6}
Evidence $\uparrow$	1.000	0.796	0.946	−0.30	9.1×10^{-6}
Label $\uparrow$	0.937	0.429	0.758	−0.66	1.1×10^{-6}

6. Conclusion

We introduce **RW-Post**, a benchmark for evidence-grounded multimodal fact-checking that aligns real-world claims with their original posts, structured reasoning traces, and explicitly linked evidence. RW-Post enables controlled evaluation of multimodal verification under complementary regimes. Experiments with strong LVLM baselines show that reliable evidence substantially improves verification performance, while visual grounding and robust misinformation refutation remain challenging. We hope RW-Post provides a useful testbed for advancing research on evidence-aware multimodal fact-checking.

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